

Problem 1:

Let X be a regular space. Show that for every pair of distinct points x, y in X , there are neighborhoods U_x, U_y of x, y with $\overline{U_x} \cap \overline{U_y} = \emptyset$.

Proof.

Let $x \neq y \in X$. Since X is regular this means $\{y\}$ is closed.

Thus, there is disjoint open sets s.t. $x \in U$ and $\{y\} \subset V$.

Thus, proven in class since X is regular there exists neighborhoods U_x, U_y s.t. $\overline{U_x} \subset U$ and $\overline{U_y} \subset V$.

Since $U \cap V = \emptyset \implies \overline{U_x} \cap \overline{U_y} = \emptyset$. ■

Problem 2:

Let $\{X_\alpha\}_{\alpha \in J}$ be a collection of nonempty spaces, and suppose that the product $\prod_{\alpha \in J} X_\alpha$ is normal. Show that X_α is normal for each $\alpha \in J$. (Note: the converse to this statement is false, but a counterexample is difficult to come by; see Exercise 9 in §32 of Munkres.)

Claim

Let $X = \prod_{\alpha \in J} X_\alpha$ be a product of topological spaces with the product topology. Let $\pi_i : X \rightarrow X_i$ be a projection map. π_i is an open map.

Proof.

Let $B = \prod U_\alpha$ be a basic open set in X s.t. U_α is open in X_α and $U_\alpha = X_\alpha$ for all but finitely many indices.

Consider $\pi_i(B)$.

If i is an index where $U_i \neq X_i$ then $\pi_i(B) = U_i$ which is open

If i is an index where $U_i = X_i$ then $\pi_i(B) = X_i$ which is open.

Thus, π_i is open on the basis sets.

Let U be open in X . Thus, $U = \bigcup B_\beta$ s.t. B_β is a basis element. Thus,

$$\pi_i(U) = \pi_i\left(\bigcup B_\beta\right) = \bigcup \pi_i(B_\beta)$$

Since $\pi_i(B_\beta)$ is open in X_i , it follows that the union of all of these which is $\pi_i(U)$ is open.

Thus, projection maps are open. ■

Proof.

Let $X = \prod_{\alpha \in J} X_\alpha$

Let X_β be one element in the collection of spaces.

Let π_β be the projection map from the product to this one space.

Let $A, B \subset X_\alpha$ be disjoint closed subsets.

Since π_β is continuous we know that $\pi_\beta^{-1}(A), \pi_\beta^{-1}(B)$ are closed in X .

Thus, since $A \cap B = \emptyset$. This implies that $\pi_\beta^{-1}(A) \cap \pi_\beta^{-1}(B) = \emptyset$. Thus, since X is normal there exists disjoint open subsets W_A, W_B in X s.t. $\pi_\beta^{-1}(A) \subset W_A$ and $\pi_\beta^{-1}(B) \subset W_B$.

Thus, by the proof above we get that $U = \pi_\beta(W_A)$ and $V = \pi_\beta(W_B)$ are disjoint open sets in X_β that contain A, B respectively.

Also, since X is normal this implies it is Hausdorff. Thus, proven in class we have shown that each component is thus Hausdorff, thus each component is also T_1 .

Thus, each component is normal. ■

Problem 3:

Show that a connected normal space containing at least two points contains an uncountable amount of points.

Proof.

Let X be a connected normal space with points $x \neq y \in X$.

By Urysohn Lemma we know there exists a continuous function $f : X \rightarrow [0,1]$ s.t. $f(x) = 0$ and $f(y) = 1$. We use the closed subsets $\{x\}$ and $\{y\}$ since X is also T_1 .

Thus, since X is connected we know that $f(X) \subset [0,1]$ is connected. Thus, by the intermediate value theorem we know since $0, 1 \in f(X)$ then $[0,1] \subset f(X)$. Thus, $f(X) = [0,1]$.

Therefore, X is uncountable since f surjects X onto an uncountable space. ■

Problem 4:

A topological space X is said to be *completely regular* if finite subsets of X are closed, and X satisfies the following condition: for every point $x \in X$ and every closed subset $A \subset X$, there exists a continuous function $f : X \rightarrow [0,1]$ such that $f(x) = 0$ and $f(a) = 1$ for every $a \in A$.

(a)

Explain why a completely regular space is always regular, and why a normal space is always completely regular

Proof.

Let X be completely regular. Let $x \in X$ and let $A \subset X$ be a closed set s.t. $x \notin A$. Then, there exists a continuous function $f : X \rightarrow [0,1]$ s.t. $f(x) = 0$ and $f(a) = 1$ for all $a \in A$.

Consider $x \in U = f^{-1}([0,1/2))$ and $A \subset V = f^{-1}((1/2,1])$ these are open sets.

Since $[0,1/2)$ is disjoint from $(1/2,1]$, This implies that $U \cap V = \emptyset$. Thus, X is regular.

Let X be normal. Then by Urysohn Lemma we know that there exists a continuous function $f : X \rightarrow [0,1]$ s.t. $f(x) = 0$ and $f(a) = 1$ for all $a \in A$.

Thus, X is completely regular. ■

(b)

Prove that every locally compact Hausdorff space is completely regular. (Use the one-point compactification.)

Proof.

Let X be a locally compact Hausdorff space. Thus, we know there exists a one point compactification $\hat{X} = X \cup \{\infty\}$.

Thus, $\hat{X}$ is a compact Hausdorff spaces proven in class we know that $\hat{X}$ is normal.

Let $x \in X$ and $A \subset X$ be a closed subset s.t. $x \notin A$. Thus, $X \setminus A$ is open in X and $\hat{X}$. Therefore, $\hat{X} \setminus (X \setminus A) = A \cup \{\infty\}$ is closed in $\hat{X}$.

Since $\hat{X}$ is normal by the problem above we get that it is completely regular thus there is a function $f : \hat{X} \rightarrow [0,1]$ s.t. $f(x) = 0$ and $f(a) = 1$ for all $a \in A \cup \{\infty\}$.

Now consider $p : X \rightarrow \hat{X}$ s.t. $p(x) = x$. Let O be open in $\hat{X}$ then if O is already open in X then $p^{-1}(O) = O$.

If $O = \hat{X} \setminus K$ for some compact set K in X then $p^{-1}(O) = X \setminus K$. Since X is Hausdorff and K is compact this means that K is closed thus $X \setminus K$ is open.

Thus, p is continuous.

Thus, consider $f \circ p : X \rightarrow [0, 1]$. This is a continuous function s.t. $f(p(x)) = f(x) = 0$, and $f(p(a)) = f(a) = 1$ for all $a \in A \subset A \cup \{\infty\}$.

Thus, X is completely regular. ■

Problem 5: Topological groups

Recall that a group G is just a set equipped with some extra structure: a binary operation satisfying certain axioms. A topological space X is also a set equipped with some extra structure: a topology $\mathcal{T}$, or a collection of subsets of X . A *topological group* is a set which is equipped with the structure of *both* a group and a topological space, in a “compatible” way. Precisely, a topological group is a group G which is also a topological space such that:

I The maps

$$G \times G \rightarrow G, \quad G \rightarrow G$$

given respectively by $(g, h) \mapsto gh$ and $g \mapsto g^{-1}$ are both continuous, and

II The singleton $\{e\}$ is a closed subset of G , where e is the identity element.

For the below, let G be a topological group.

(a)

Prove that any topological group satisfies the T_1 axiom.

Claim

Let X, Y be topological spaces then $f : X \rightarrow X \times \{y\}$ for $y \in Y$ is a homeomorphism.

Proof.

Injective:

$$\text{Let } a, b \in X \text{ s.t. } f(a) = f(b) \implies (a, y) = (b, y) \implies a = b$$

Surjective:

$$\text{Let } (a, y) \in X \times \{y\}. \text{ Consider } f(a) = (a, y)$$

Continuous:

Let U be a basic open set in $X \times \{y\}$. Thus, $U = V \times W$ for V open in X and W open in $\{y\}$.

$$\text{Thus, } f^{-1}(U) = V.$$

Continuous Inverse:

Let U be an open set in X . Thus, $f(U) = U \times \{y\}$. This is a basic open set in $X \times \{y\}$.

Thus, $f(U)$ is open.

Therefore, f is a homeomorphism. ■

Proof.

Let $m : G \times G \rightarrow G$ be the binary operation map.

Define: $f_x : G \rightarrow \{x\} \times G$ s.t. $f_x(g) = (x, g)$

Define: $L_x : G \rightarrow G$ s.t. $L_x = m \circ f_x$

Injective:

Let $a, b \in G$ s.t. $L_x(a) = L_x(b) \implies xa = xb$. Multiply by x^{-1} on both sides, and we get that $a = b$.

Surjective:

Let $a \in G$ consider $y = x^{-1}a$

Thus, $L_x(y) = xx^{-1}a = a$

Continuous:

L_x is continuous since f_x by the claim and proof above and m is continuous by definition.

Continuous Inverse:

Since the inverse of $L_x^{-1} = L_{x^{-1}}$ is also continuous by the same proof above we get that L_x is a homeomorphism.

Therefore, we get that $L_x(\{e\}) = \{x\}$ is closed. ■

(b)

Prove that any topological group is Hausdorff.

Proof.

Let $i : G \rightarrow G$ be the inverse map.

Define: $f : G \times G \rightarrow G$ s.t. $f(g, h) = m(g, i(h)) = gh^{-1}$. This is continuous since m, i and the identity map are continuous.

Now consider $f^{-1}(\{e\}) = \{(g, h) \in G \times G \mid gh^{-1} = e\}$. Since inverse are unique in groups this is only true when $g = h$.

Thus, we have shown that $\Delta = \{(g, g) \mid g \in G\}$ is closed. Thus, by previous homework G is Hausdorff. ■

(c)

Prove that any topological group is regular.

Proof.

Consider the continuous map from the last problem $f(g, h) = gh^{-1}$.

Let A be a closed set s.t. $e \notin A$.

Consider $f^{-1}(X \setminus A)$. This is an open set in $G \times G$ since f is continuous.

Since $f(e, e) = e \in X \setminus A \implies (e, e) \in f^{-1}(X \setminus A)$. Therefore, let $V_1 \times V_2$ be a basic open

set s.t. $(e, e) \in V_1 \times V_2 \subset f^{-1}(X \setminus A)$.

Define: $V = V_1 \cap V_2$. This is open since V_1, V_2 are open in G .

Consider $AV = \bigcup_{a \in A} L_a(V)$. This is an open set since $L_a(V)$ is open since L_a is an homeomorphism.

Let $a \in A$ since $e \in V \implies ae = a \in AV$. Thus, $A \subset V$.

Let $x \in V$ and $x \in AV$. Thus, $x = ay$ for some $a \in A$ and $y \in V$. Thus, $a = xy^{-1}$. This is a contradiction since $x, y \in V \implies f(x, y) \in X \setminus A \implies a \in X \setminus A$.

Therefore, $V \cap AV = \emptyset$, and $e \in V, A \subset AV$.

To do this for any $x \in G$ and any closed set C s.t. $x \notin C$. Just do this for e , and $x^{-1}C$. $e \notin x^{-1}C$ since if it was then this would imply that $x \in C$ since inverses are unique.

Thus, we get disjoint open sets s.t. $e \in V$ and $x^{-1}C \subset U$.

Then consider $x \in xV$ and $C \subset xU$. These are still disjoint since if there were not this would imply that $V \cap U \neq \emptyset$ since left multiplication is a homeomorphism.

Thus, G is regular. ■